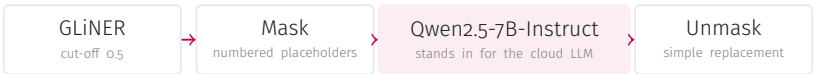
Toy Experiment: Setup



CLEAN Write a short message to my landlord. I'm Minji Park 0.999 from 41 Maple Street 0.969, and my heater is broken. He can call me at 010-2345-6789 1.000.

NOISY write a short message to my landlord im minji park 0.686 from 41 maple stret 0.921 and my heater is broken he can call me at 010 2345 6789 0.985

GLiNER masks all three values in both prompts, but the lowercase name drops from 0.999 to 0.686, close to the 0.5 cut-off.

Highlight = span GLiNER masked; number = its confidence. Detector: urchade/gliner_multi_pii-v1. LLM: Qwen/Qwen2.5-7B-Instruct, run locally with greedy decoding.

Toy Experiment: Example 1 (Clean)

YOU WRITE Write a short message to my landlord. I'm Minji Park from 41 Maple Street, and my heater is broken. He can call me at 010-2345-6789.

LLM RECEIVES Write a short message to my landlord. I'm <NAME_1> from <ADDRESS_1>, and my heater is broken. He can call me at <PHONE_1>.

LLM ANSWERS Dear Landlord, … My name is <NAME_1> and I live at <ADDRESS_1>. … call me at <PHONE_1> … Best regards, <NAME_1 ← **no ">"**

YOU READ Dear Landlord, … My name is Minji Park and I live at 41 Maple Street. … call me at 010-2345-6789 … Best regards, <NAME_1 ← **not restored**

"…" marks text cut for the slide; the full answer is in results/toy/local/toy_results.md.

Toy Experiment: Results

	Example 1 (clean)	Example 2 (noisy)
PRIVATE?	**Yes** no real value sent	**Yes** no real value sent
ENOUGH CONTEXT?	**Yes** all 3 placeholders used	**Yes** all 3 placeholders used
USABLE ANSWER?	**No** sign-off left as `<NAME_1`	**No** same, and the typos come back ("41 maple stret")

- **The loop works:** the LLM wrote a usable message without seeing any real value.
- **Unmasking is the weak step:** the LLM dropped the closing ">" of its last placeholder, which simple replacement cannot restore.
- **RQ2 hint:** typing the name in lowercase cut its confidence from 0.999 to 0.686; nothing leaked, but close to the cut-off.